HEALTHCARE QI (PDSA / QCC) & MEDICAL PRESENTATION REDESIGN

From Messy Clinical Notes & Raw Excel Tables -> 100% Native Editable Executive PowerPoint

SAMPLE — FICTIONAL DATA

BEFORE: Typical Cluttered Clinical Draft

ICU CAUTI Quality Project Q1-Q2 Update

• Baseline CAUTI rate was 6.8 per 1000 catheter days in Jan-Feb which is above benchmark of 2.5.
• We audited 130 defects: missed daily review 50, tubing loop 35, aseptic breach 24, unsecured 12, hygiene 9.
• Root causes: no EHR reminder, float nurses unfamiliar with protocol, clips out of stock in supply room.
• After implementing nurse-driven removal protocol and green clips, rate dropped to 1.9 per 1000 days.
• Compliance went from 64% up to 96.5% and estimated savings around $148,500 annually.

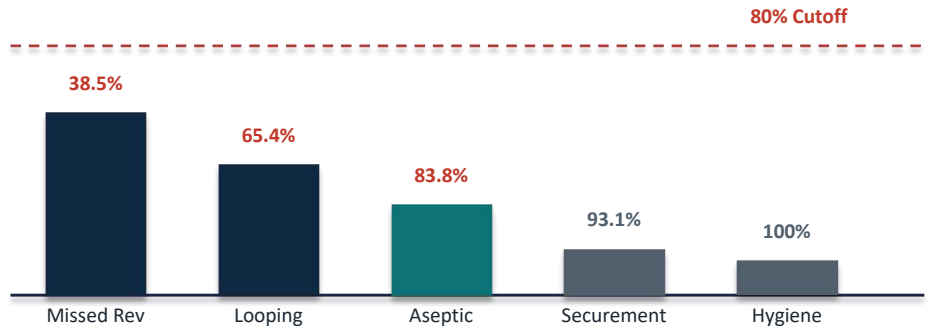
- Dense text wall - No visual 80/20 Pareto hierarchy
- Pasted flat screenshots that cannot be edited last-minute

PRO FIX

AFTER: Executive Board & Conference Ready (.PPTX)

CAUTI Rate	Bundle Audit	Cost Avoided
6.8 -> 1.9	64% -> 96.5%	$148.5K
-72.1% (p<.001)	+32.5 pts	Annualized

80/20 Pareto Vital Few: Top 3 Causes Drive 83.8% of ICU Defects (n=130)



+ 100% Editable Shapes, Pareto, Fishbone & SPC Run Charts
+ Zero Font Shift on Windows & Mac Office 365 | Ready for Magnet / ICQCC / Posters

- IHI PDSA & Lean A3 Ready
- Editable Pareto & 6M Fishbone
- 48"x36" Scientific Posters
- Office 365 Safe Typography

IHI PDSA QUALITY IMPROVEMENT · MAGNET & ICQCC STANDARD

Reducing Catheter-Associated Urinary Tract Infections (CAUTI) in the Adult ICU

Sample — Fictional Data · 24-Week Multidisciplinary Nurse-Driven Bundle & EHR Initiative

CLINICAL SETTING
28-Bed Adult Med-Surg ICU
St. Jude Metropolitan Medical Center

PROJECT WINDOW
24 Weeks (Baseline + 3 PDSA Cycles)
1,840 Audited Catheter-Days

QI CIRCLE LEADERS
Sarah Jenkins, MSN, RN, CCRN (Lead)
David Chen, MD, MPH (Advisor)

KEY OUTCOME SNAPSHOT

-72.1%
CAUTI Rate Reduction
(6.8 -> 1.9 / 1,000 days)

96.5%
Post-PDSA Daily Bundle
Compliance (+32.5 pts)

$148.5K
Annualized Direct ICU
Cost Avoidance

Executive Summary & SMART Quality Improvement Charter

PRIMARY OUTCOME · CAUTI RATE

1.9

per 1,000 catheter-days

Baseline: 6.8 | Target: <=2.5 (-72.1%, p<0.001)

DEVICE UTILIZATION RATIO

0.51

catheter-days / patient-days

Baseline: 0.74 | Target: <=0.55 (-31.1%)

BUNDLE AUDIT COMPLIANCE

96.5%

all-or-none 5-point check

Baseline: 64.0% | Target: >=90.0% (+32.5 pts)

ESTIMATED COST AVOIDANCE

$148,500

annualized ICU direct cost

11 prevented CAUTI cases x $13,500 AHRQ unit cost

Background & IHI SMART Aim Statement

Clinical Gap: In Q4 baseline surveillance, our 28-bed Adult ICU recorded 6.8 CAUTIs per 1,000 catheter-days (95th percentile NHSN outlier), adding 3.8 excess ICU days per event.

SMART Aim: Reduce Adult ICU CAUTI incidence from 6.8 to <=2.5 per 1,000 catheter-days (-63% target) and raise daily catheter-care bundle compliance from 64.0% to >=90.0% within 24 weeks.

Methodology: 10-Step Healthcare QCC / 3-Cycle IHI PDSA framework anchored on CDC HICPAC Guidelines (Category IB) and direct bedside observation (n = 420 catheter-shift audits).

Multidisciplinary QI Circle Composition (7 Members)

Role	Credential / Title	QI Responsibility
Circle Leader	Sarah Jenkins, MSN, RN, CCRN	Protocol governance & weekly PDSA huddle
Physician Advisor	David Chen, MD, MPH	Medical staff alignment & indication criteria
Infection Control	Elena Rostova, MPH, CIC	NHSN CAUTI adjudication & SPC surveillance
EHR Analyst	Marcus Vance, BSN, RN-BC	24h Epic hard-stop order set & alert build
Bedside Champions	3 Senior ICU Charge RNs	Shift peer audits & supply par-level check

Baseline Bedside Defect Audit & 80/20 Pareto Vital Few Analysis

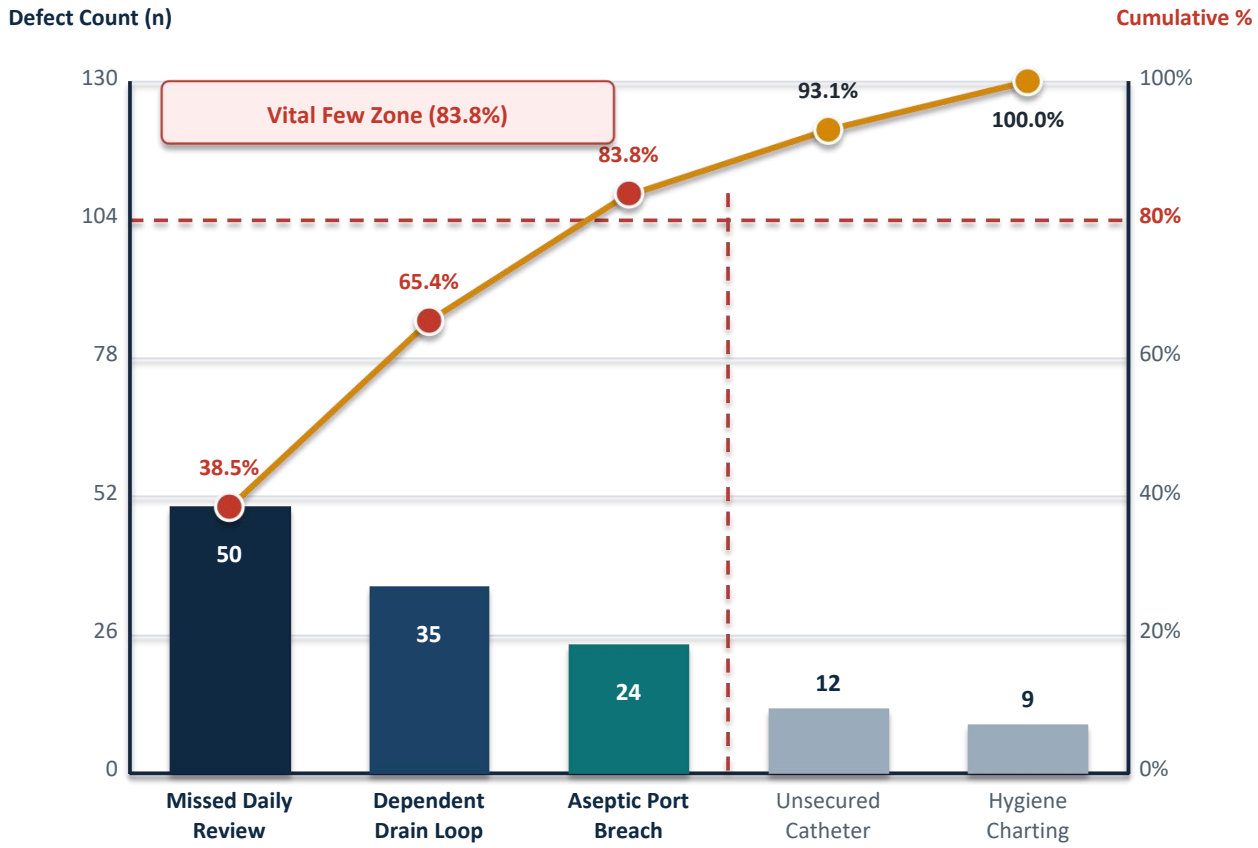
n = 130 Observed Defects (420 Audits)

Baseline Defect Stratification Table (Jan–Feb)

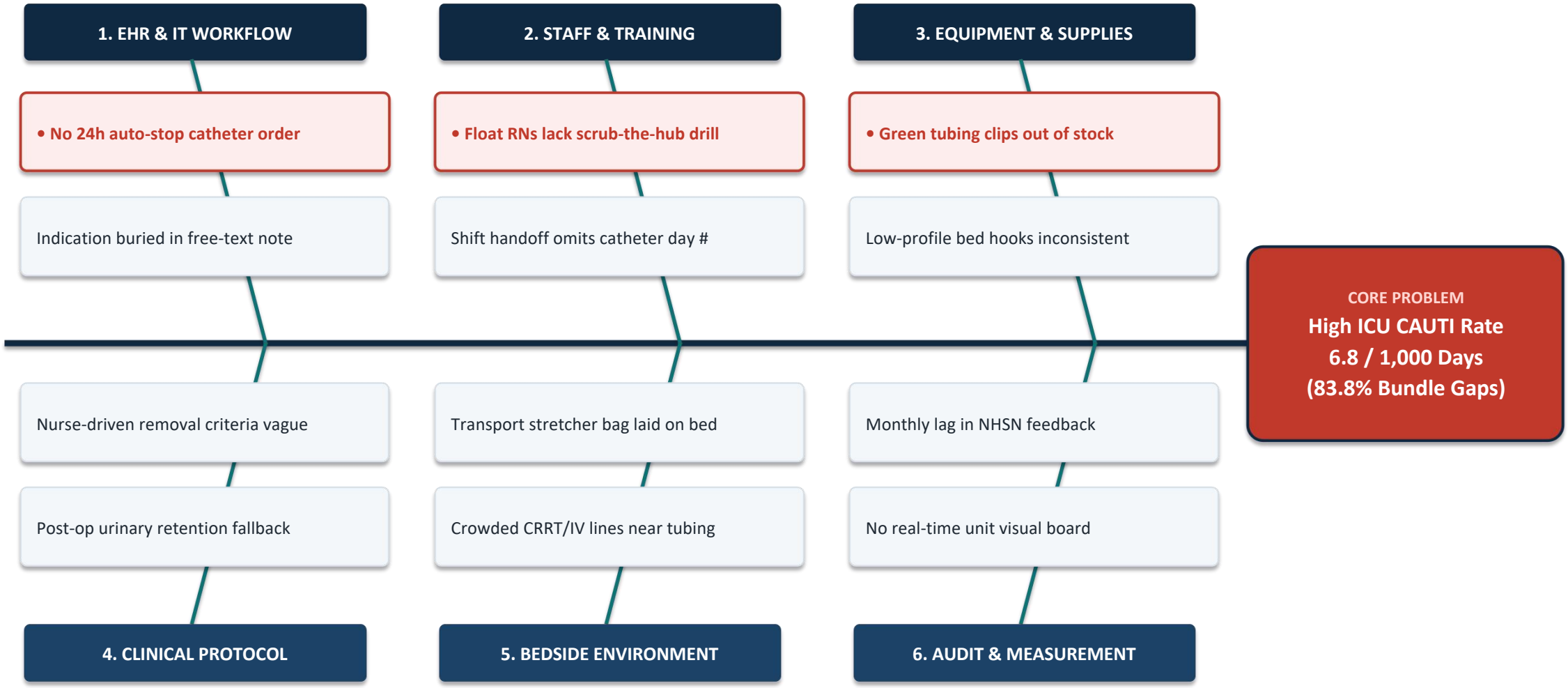
Rank & Defect Category	Defects	Share	Cum %
1. Missed Daily Necessity Review •	50	38.5%	38.5%
2. Dependent Tubing Loop •	35	26.9%	65.4%
3. Aseptic Port Maintenance Breach •	24	18.5%	83.8%
4. Unsecured Thigh Securement	12	9.2%	93.1%
5. Incomplete Hygiene Charting	9	6.9%	100.0%
Total Audited Bundle Defects	**130**	**100.0%**	—

80/20 Pareto Conclusion (Vital Few Focus):
Top 3 failure modes account for 109 of 130 total defects (83.8%) and were selected as primary PDSA intervention targets.

Pareto Chart of Baseline CAUTI Bundle Failure Modes (n = 130)



6M Ishikawa Fishbone Analysis & Verified True Root Causes



• Verified True Root Causes ($p < 0.01$ on bedside validation): [RC-1] Missing 24h EHR hard-stop review | [RC-2] Float RN aseptic port gap | [RC-3] Drainage clip stockout

3-Cycle IHI PDSA Ramp & 5W2H Action Implementation Matrix

PDSA Cycle	Target True Cause	Intervention Bundle (What & How)	Owner (Who)	Window	Verified Result
PDSA Cycle 1 (EHR System)	RC-1: No 24h catheter necessity prompt	Deploy Epic 24h mandatory indication banner + Nurse-Driven Foley Removal Protocol (HOUDINI criteria) with bladder-scan algorithm.	EHR Analyst & ICU Medical Dir.	Weeks 5–10	**Missed daily review dropped 38.5% -> 4.2%; DUR 0.74 -> 0.55**
PDSA Cycle 2 (Poka-Yoke Hardware)	RC-3: Dependent tubing loops & clip stockout	Install color-coded Green-Zone Bedrail Tubing Clips on all 28 ICU beds + add clip to Central Supply Foley insertion kit par level.	ICU Supply Lead & Charge RNs	Weeks 9–15	**Dependent loop defects fell 35 -> 3 (-91.4% reduction)**
PDSA Cycle 3 (Competency & Huddle)	RC-2: Float RN aseptic port maintenance gap	15-sec 'Scrub-the-Hub' UV-glow competency checkoff for all ICU/float RNs + add 'Foley Day #' to bedside SBAR shift handoff card.	Clinical Nurse Educator & Champions	Weeks 13–20	**All-or-none bundle compliance rose 64.0% -> 96.5%**

CYCLE 1 · PLAN-DO-STUDY-ACT
Test HOUDINI removal protocol on 8-bed Pod A -> refine post-op neuro exclusion -> scale to all 28 beds.

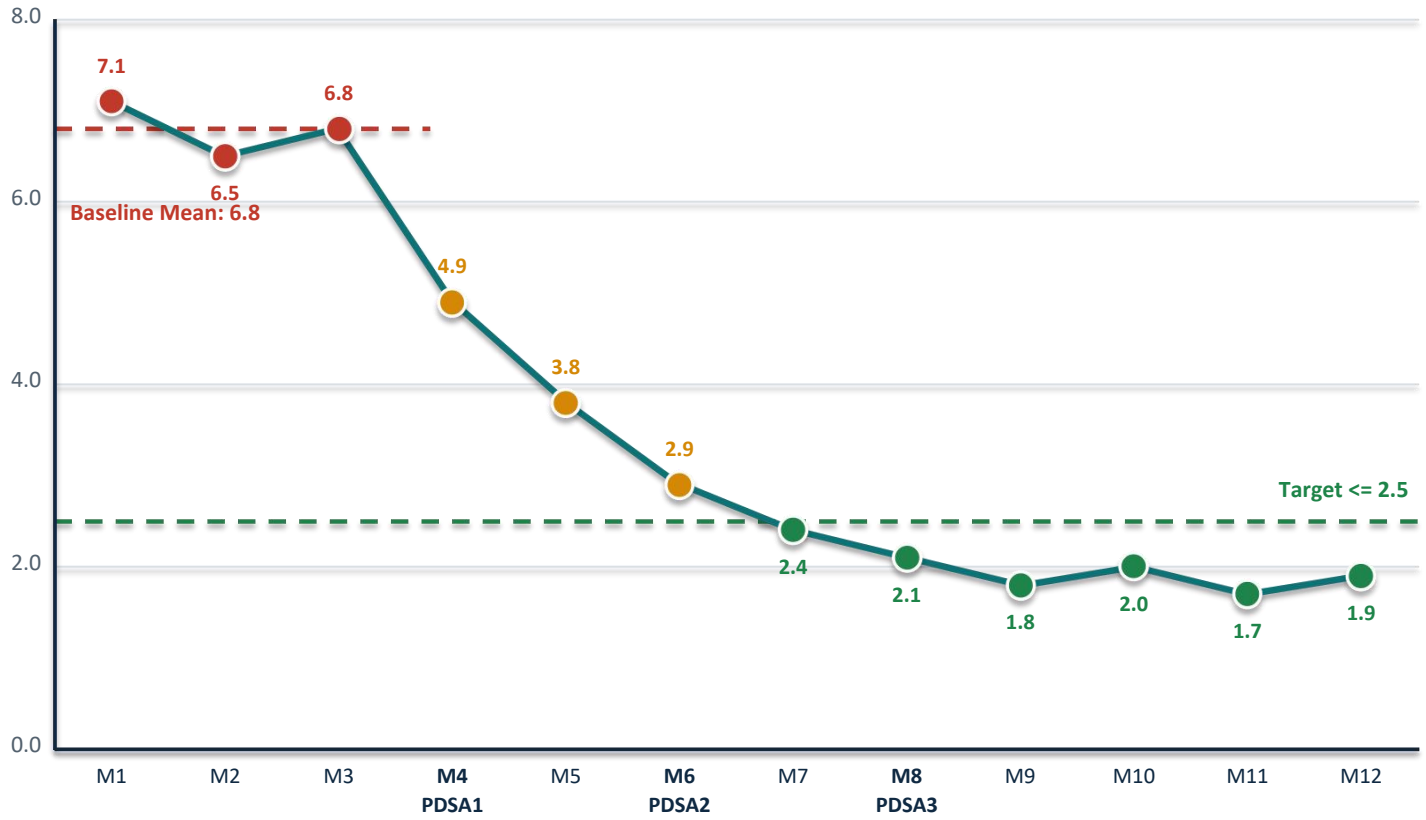
CYCLE 2 · POKA-YOKE HARDWARE
Pilot 3 tubing clip models during prone/transport turns -> lock non-slip green rail clamp across ICU & ER.

CYCLE 3 · SUSTAIN & AUDIT
Embed weekly 10-chart peer audit into Charge RN tablet checklist -> post unit 'Days Since Last CAUTI' counter.

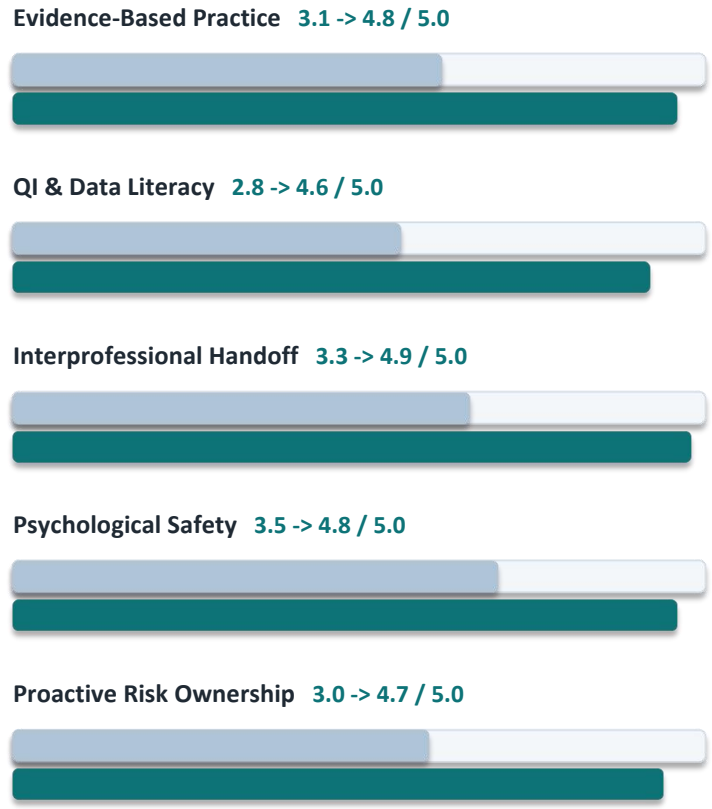
Longitudinal Outcome: 12-Month CAUTI SPC Run Chart & Team Gains

Pre vs. Post p < 0.001

Monthly ICU CAUTI Rate per 1,000 Catheter-Days (12-Month Surveillance)



Intangible Team Capability Gains (5-Point Validated Survey, n=42 RNs)



Goal Achievement Rate: 114.0% | Progress: +72.1%

Standardized ICU CAUTI Prevention Bundle & Sustainability Control Plan

GATE 01 · INSERTION

Strict CDC Indication Gate

- Hard-stop order menu (6 approved indications only)
- Bladder ultrasound >=350 mL check prior to catheterization
- 2-RN aseptic insertion witness checklist

GATE 02 · 24H EHR REVIEW

Automated Daily Prompt

- 07:00 Epic banner fires at >=24h dwell time
- RN autonomously executes HOUDINI removal if no exclusion
- Physician override expires every 24 hours

GATE 03 · BEDSIDE CARE

Poka-Yoke Physical Lock

- Green-Zone bedrail clip keeps bag below bladder & off floor
- 15-sec chlorhexidine/alcohol port scrub before sampling
- 'Foley Day #' spoken at every SBAR shift handoff

GATE 04 · GOVERNANCE

Closed-Loop SPC Audit

- Weekly 10-bed random Charge RN tablet audit
- Real-time unit safety huddle board ('47 Days CAUTI-Free')
- New-hire & float RN simulation onboarding sign-off

DOCUMENT CONTROL

Code: SOP-ICU-CAUTI-V3
Approved: Nursing Quality Council
Horizontal Spread: CCU, Neuro-ICU & Stepdown (Q3)

Key Clinical References (AMA Format):

1. Gould CV, Umscheid CA, Agarwal RK, et al. Guideline for prevention of catheter-associated urinary tract infections. Infect Control Hosp Epidemiol. 2010;31(4):319-364.
2. Patel PK, Advani SD, Kofman AD, et al. Strategies to prevent catheter-associated urinary tract infections in acute-care hospitals: 2023 update. Infect Control Hosp Epidemiol. 2023;44(8):1209-1243.

48" x 36" Scientific & Medical Conference Poster Layout (Editable PPTX)

Reducing Adult ICU Catheter-Associated UTIs via EHR Decision-Support & Nurse-Driven PDSA Bundles

Sarah Jenkins, MSN, RN, CCRN; Elena Rostova, MPH, CIC; Marcus Vance, BSN, RN-BC; David Chen, MD, MPH | St. Jude Metropolitan Medical Center

48" x 36" POSTER
100% EDITABLE

1. BACKGROUND & SMART AIM

- Baseline ICU CAUTI rate reached 6.8 / 1,000 catheter-days (NHSN >90th percentile).
- Aim: Cut CAUTI <=2.5 / 1,000 days and achieve >=90% bundle compliance within 24 weeks.
- Setting: 28-bed Adult Med-Surg ICU (1,840 catheter-days).

2. BASELINE PARETO VITAL FEW (n=130)

- **Missed Daily Review: 50 (38.5%)**
- **Dependent Drain Loop: 35 (26.9% | Cum 65.4%)**
- **Aseptic Port Breach: 24 (18.5% | Cum 83.8%)**
- Unsecured / Hygiene: 21 (16.2% | Cum 100%)

-> Top 3 vital few defects targeted via 6M RCA.

3. MULTIDISCIPLINARY PDSA INTERVENTIONS

PDSA 1 · EHR 24h Hard-Stop & HOUDINI
Automated Epic dwell alert + autonomous RN removal protocol with bladder scan algorithm.

PDSA 2 · Poka-Yoke Green Tubing Clips
Standardized non-slip bedrail clips installed across all 28 beds + kit par-level lock.

PDSA 3 · Scrub-the-Hub & SBAR Handoff
UV-glow aseptic port checkoff for all ICU/float RNs + daily catheter day # in handoff.

4. CLINICAL & FINANCIAL OUTCOMES

- **CAUTI Incidence: 6.8 -> 1.9 / 1,000 days (-72.1%, p < 0.001)**
- **Device Utilization: 0.74 -> 0.51 (-31.1%)**
- **Bundle Compliance: 64.0% -> 96.5% (+32.5 pts)**
- **Direct Cost Avoidance: $148,500 annualized (11 prevented cases)**

5. SUSTAINABILITY & CONCLUSIONS

Pairing EHR forcing functions with physical poka-yoke securement yielded sustained sub-benchmark CAUTI rates across 6 months of follow-up. Protocol SOP-ICU-CAUTI-V3 scaled to CCU & Neuro-ICU.